

Q/R/M/C: A Stage-Decomposition Evaluation Protocol for Memory-Based Agents

Anonymous submission

Abstract

Success-only metrics collapse qualitatively different memory failures into a single scalar and hide which stage of the retrieval pipeline is responsible. We describe an evaluation protocol, the Q/R/M/C decomposition (Query formation, Retrieval satisfaction, target Materialization, Completion), operationalised so that four conditional rates can be computed from standard trajectory logs. The arithmetic is an elementary chain rule over logged events; the substance is in the event definitions, which make the rates estimable in practice, much as precision and recall are useful because of their definitions and not their arithmetic. In single-agent experiments (MiniGrid, BabyAI), oracle-stage interventions separate a retrieval-limited task ($0.39 \rightarrow 0.60$ under oracle retrieval) from downstream-limited ones, and log inspection refines the diagnosis: ranking among retrieved candidates is near-optimal (96.9%), but the candidate set is empty at 91% of decision points, so the limiting factor is memory accumulation. In multi-agent experiments (35,640 runs, four sharing architectures, cadence K), the predicted bottleneck shift between Materialization and Completion is confirmed as cluster-robust slopes (Spearman $-0.53/+0.43$), time-to-success among successes has an interior optimum at $K=8$, and a provenance annotation identifies the mechanism: under fast sharing the completion collapse is concentrated in locks backed by peers' evidence. The same tool-trace contract also instruments three third-party agent stacks (OpenAI SDK, LangGraph, AutoGen). Engineered structural failures localise identically on every stack ($R^*=0$, $C^*=0$ in 20/20), while on clean tasks the frameworks spread 0.50–0.75 end-to-end, and the stages attribute the spread to weak first-commitment discipline combined with different recovery loops. The protocol makes memory failures comparable across symbolic and LLM systems, single- and multi-agent, wherever a query/lock interface is exposed; on opaque learned policies it measures whether the stages are observable at all.

Introduction

Embodied, goal-conditioned, and LLM agents increasingly retrieve places, objects, and facts *by meaning* from an explicit memory, yet they are evaluated almost exclusively end-to-end: if success improves, memory is judged useful. Two

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systems with the same success rate can fail for different reasons. One never accumulates queryable evidence; another retrieves perfectly and then loses the target to a peer. This paper describes a measurement instrument that separates such cases.

The instrument is a four-stage decomposition of memory-based task completion: **Query** formation (the planner emits a semantic intent), **Retrieval** satisfaction (memory returns a candidate matching the task), target **Materialization** (the planner locks one candidate as an actionable target), and **Completion** (the controller reaches it). Logged as per-episode events Q^* , R^* , M^* , C^* , they factorise semantic-path completion into conditional rates estimable from ordinary trajectory logs (Definition 1; a multi-agent extension with memory and occupancy coupling channels is Definition 2). The consistency argument is the chain rule plus frequency estimation, and we claim no mathematical novelty for it. The precedent is the decomposition of accuracy into precision and recall, which was also arithmetically trivial and still changed evaluation practice. Our contribution is a logging discipline and the diagnoses it supports.

Contributions.

1. **The protocol.** Operational, log-level definitions of the four stage events for single agents (Definition 1) and communicating collectives (Definition 2), with identifiability under a stated conditional stage-Markov assumption, plus a stress test of that assumption.
2. **Stage-level diagnoses.** Single-agent: oracle-stage interventions and log forensics attribute failures to memory accumulation rather than to ranking. Multi-agent: a registered slope-level claim (Empirical Claim 1: fast sharing raises $P(M^*|R^*)$ and lowers $P(C^*|M^*)$) is confirmed on a 35,640-run sweep, with an interior time-to-success optimum at $K=8$ and a provenance-level mechanism.
3. **External validity.** A two-layer contract (invariant event roles plus a system-specific adapter) instruments OpenAI SDK, LangGraph, and AutoGen agents with one shared tool-trace adapter and no per-framework changes; engineered structural failures localise identically on all stacks, and behavioural differences between stacks are attributed stage-by-stage.

Table 1: Portability of the contract across system classes: the same four events, defined once, instrument each class without modification and localize a different failure in each. Sections – instantiate the rows in order.

System class	What the same four events revealed
Symbolic single agent	ranking 96.9% vs. empty candidates 91%: an <i>accumulation</i> limit
Symbolic collective	$M \leftrightarrow C$ bottleneck shift; t_{succ} optimum $K=8$; provenance mechanism
Learned policies	competitive success, collapsed stage observability
LLM agents	$1.00 \rightarrow 0.00$ collapse = pure C-budget; sharing lowers success $2/3 \rightarrow 1/3$
Third-party stacks $\times 2$ models	structural failures localize identically; M_1 separates models; frameworks reshape it

Scope and non-claims. Each limitation is stated together with the reason the results survive it. (i) The contribution is the operationalisation: log-level event definitions under which the four conditional rates are identifiable on any system exposing a query/lock interface. The factorisation arithmetic itself (the chain rule) carries no mathematical novelty, and none is claimed. (ii) Empirical Claim 1 is a slope-level empirical observation rather than a theorem, stated deliberately so that each slope is separately falsifiable on logs; both were confirmed with cluster-robust confidence intervals. (iii) No claim is made about the conditional product Φ , which is monotone in our data; the interior optimum lives on conditional time-to-success, and we report the censoring-aware contrast under which it does not persist. The operationalisation is testable on specific summaries and can be wrong on others; a protocol without such exposure would be unfalsifiable. (iv) Primary substrates are grid worlds because they admit oracle interventions and exhaustive sweeps (35,640 runs) that richer substrates do not; the directional VMAS check and the LLM and third-party studies indicate that the event definitions do not depend on grid structure. (v) Multi-agent place identity assumes a shared coordinate frame in the main sweep; a companion manuscript removes the assumption constructively, leaving the event definitions unchanged. (vi) R^* and M^* presuppose an entity correspondence between memory content and world referents, supplied here by instrumented environments. The requirement is mild in the common evaluation setting of this paper — interchangeable agents (a shared perception interface, possibly different weights) situated in a single environment inherit the correspondence from the shared substrate — and becomes a genuinely open problem only for heterogeneous agents in open worlds, where the protocol degrades gracefully to Q^* and C^* .

The Q/R/M/C Protocol

An episode issues a semantic query (required/preferred/penalty tags); memory returns candidates; the planner locks one; the controller executes. The four events anchor at logged triggers on the episode’s query–lock chain and are *nested by construction*: Q^* fires on intent emission; $R^* := Q^* \wedge$ (the returned candidate set contains a task-satisfying candidate); $M^* := R^* \wedge$ (the planner locks a true target drawn from that candidate set); $C^* := M^* \wedge$ (the controller reaches the locked target). An episode may run several query–retrieve–lock attempts; a link fires if

some attempt realizes it together with its predecessors, so $C^* \subseteq M^* \subseteq R^* \subseteq Q^*$ holds at the episode level by definition. C^* is *semantic-path completion* — completion through the retrieval path — and is distinct from raw task success, which can also occur off-path.

We keep two kinds of quantities separate. *Link rates* are the conditionals $q = P(Q^*)$, $r = P(R^* | Q^*)$, $m = P(M^* | R^*)$, $c = P(C^* | M^*)$, estimated by conditional frequency over episodes; under the nesting above, $P(C^*) = q r m c$ is an exact identity — a chain rule that needs no further assumption. *Marginal event frequencies* are used as four separate diagnostics on systems whose logs expose no lock chain (§); there an event can occur without its predecessor, the events are not nested, and no product is claimed.

Definition 1 (single-agent estimability, informal). With the nested events above, $P(C^*) = q r m c$ holds identically and each link rate is consistently estimable from logs. The conditional stage-Markov property (each stage outcome depends on the trace only through its predecessor stage) is not needed for the identity; it is what licenses reading each link as a stable property of the owning subsystem, and hence predicting the effect of a stage intervention. A simulation stress test quantifies what breaks when it fails: with a hidden confounder, m stays unbiased while c inflates, so diagnoses that hinge on the M – C contrast degrade gracefully (supplement).

Definition 2 (multi-agent extension, informal). With N agents sharing memory by peer broadcast every K ticks, the conditioning sets enlarge with two coupling channels: the shared memory $\mathcal{M}_{-i}^{(K)}$ and occupancy $\mathcal{O}_{-i}$ (scarce targets are consumed). The same four rates are logged per agent.

Empirical Claim 1 (bottleneck shift, slope-level). As K decreases (faster sharing), $P(M^* | R^*)$ rises, because fresher peer evidence enables reliable commitment, and $P(C^* | M^*)$ falls, because peers race for the same targets. Both are claims about conditional rates, not about their product.

Interpretive readings of the stages (an order-theoretic account of empty retrieval via the query–extent Galois connection; an options/semi-MDP account of why optima appear on time-bearing metrics) are confined to the supplement and are not counted among the contributions.

Single Agent: Localizing the Failure

On a 10-seed MiniGrid benchmark (water/rest semantic search, goal-region rejoin in FourRooms, hazard recovery in LavaGapS7; BabyAI portability in the supplement), end-to-end success hides heterogeneity: water and rest are strong (0.95, 0.93), while rejoin and hazard recovery are weak. The reasons differ, and oracle-stage interventions separate them. Replacing retrieval with an oracle lifts hazard-recovery success $0.39 \rightarrow 0.60$ (bootstrap 95% CIs $[0.35, 0.43]$ vs. $[0.51, 0.69]$; per-seed paired deltas positive in 9/10 seeds, exact sign test $p=0.004$; retrieval-limited); the same oracle rescues semantic-path completion on goal-region rejoin without changing end-to-end success (downstream-limited). The framework’s own logs refine the attribution further: when candidates exist, ranking precision is 96.9%, but the candidate set is empty at 91% of decision points — a figure

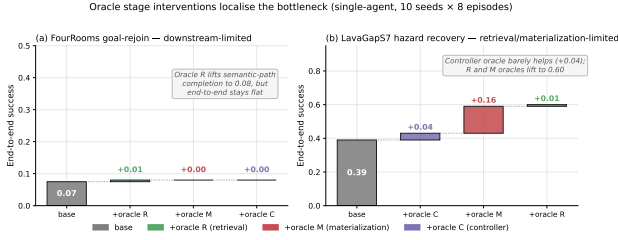


Figure 1: The diagnose→intervene→verify loop on the single-agent benchmark: oracle-stage interventions lift exactly the stage that the decomposition identifies. Hazard recovery is retrieval-limited (oracle-R: 0.39 → 0.60 end-to-end); goal-region rejoin is downstream-limited (oracle-R rescues semantic-path completion, end-to-end unchanged).

robust to the weighting (0.908 pooled over calls, 0.909 seed-weighted, 0.891 episode-weighted), so it is not an artefact of long failing episodes contributing more decision points. The binding constraint is accumulation, since the agent rarely has anything to rank. In our own development this number redirected effort from rankers to consolidation; success-only evaluation gives no indication of it. Accumulation itself is not one of the four stages: the protocol localises the failure to the R link, and within-link log forensics names the cause. We therefore read Q/R/M/C as coarse localisation with a designated drill-down, not as a complete causal ontology of memory failure.

A learned behaviour-cloning baseline illustrates the observability caveat: it reaches competitive success with collapsed stage structure (there is no explicit query/lock interface), so end-to-end optimisation removes exactly the events the protocol needs.

Multi-Agent: Memory as a Shared Resource

Four sharing architectures run a scarce water-search task over one symbolic substrate: independent private memories, a shared event bus, a central trust-weighted aggregator, and peer-to-peer broadcast at cadence $K \in \{1, \dots, 64\}$ ($N \in \{3, 5, 8\}$ agents, $T \leq N$ targets, three layouts, three hazard densities, 40 seeds; 35,640 runs). Figure 2 shows the stage profile of every architecture; retrieval saturates, and the differences between architectures are confined to the M and C links, as Empirical Claim 1 predicts.

Bottleneck shift. Across peer runs, Spearman of $P(M^*|R^*)$ vs. K is -0.53 (cluster-robust 95% CI $[-0.59, -0.47]$ over 81 configuration cells) and of $P(C^*|M^*)$ vs. K is $+0.43$ ($[+0.37, +0.50]$). The sharper signature is within-cadence: Pearson between the two links inside a fixed K peaks at -0.61 at $K=4$ and decays to noise by $K=64$, so the bottleneck migrates between stages within the same configuration as K varies. Scaling to $N=12$ sharpens the signature (peak -0.70 , migrating to $K=8$). On mean time-to-success among successful runs the trade-off produces an interior optimum at $K=8$ (7.63 vs. 8.04 independent, cell-weighted): cell-paired contrasts show $K=8$ faster than $K=16$ (-0.39 , 95% CI $[-0.80, -0.08]$) and than

independent (-0.40 $[-0.59, -0.24]$), and statistically tied with $K \in \{2, 4\}$. That metric conditions on success. Under a censoring-aware alternative that charges failed runs the full step limit, the fast end reverses and independent is marginally best (paired $\Delta(K=8 - \text{indep}) = +0.32$ $[+0.05, +0.63]$): fast sharing’s completion collapse costs more successes than its materialization gain buys back. The interior optimum is therefore a statement about conditional speed, not unconditional efficiency; consolidation rate remains a tunable design parameter, and closing the unconditional gap is what the reservation protocol below achieves. Oracle interventions confirm the attribution: at fast cadence, oracle-R/M/C progressively remove the excess time, placing it at the $M \leftrightarrow C$ transition.

Provenance stratification. We annotate each M^* with the foreign share φ of its supporting evidence mass (computed from the merge’s own weights, frozen at lock time) and call locks with $\varphi \geq 0.8$ warps. The warp share falls from 0.55 at $K=1$ to 0.11 at $K=8$, while own-evidence locks complete at a flat 0.26–0.34 and warp locks stay below 0.05 throughout. The completion collapse of fast sharing is therefore concentrated in the foreign-evidence stratum: cadence controls how much of the collective’s commitment relies on peers’ evidence, and this stratum loses the occupancy race. The stratification is not an artefact of the 0.8 cutoff: recomputed at thresholds $\varphi \in \{0.5, \dots, 0.95\}$, the foreign stratum completes at ≤ 0.03 while the own stratum stays at 0.26–0.50, at every threshold and cadence. The two coupling channels also separate interventionally with arms already in the design: the independent arm removes peer evidence while leaving target contention in place, and a decentralized reservation-and-rollback protocol (companion manuscript) keeps peer evidence while managing contention — under it, unconditional success at fast cadence exceeds the $K=8$ baseline (0.614–0.620 vs. 0.583), identifying occupancy racing rather than evidence quality as the cost channel of fast sharing. The same stratification curve reproduces on a continuous VMAS bridge (0.72 → 0.00) and on an LLM text substrate (0.42 → 0.08).

Learned and LLM collectives. A CommNet-PPO baseline matches symbolic peer success (0.667 vs. 0.65) with structurally degenerate stages: its real-valued channel is never informatively empty, so R and M collapse and the stages cannot be read off at competitive success. A collective of three LLM agents (llama3.1) sharing symbolic memory reproduces the phenomenology on language. Paired against a no-communication baseline on identical layouts, raw sharing lowers success $2/3 \rightarrow 1/3$ (occupancy racing), and a decentralized reservation-and-rollback protocol recovers part of it (0.33 → 0.42) with rollback latency $\approx K$. On a step-budget sweep with two local models, an opaque success collapse ($1.00 \rightarrow 0.00$) resolves to pure C-stage failure ($Q^*=R^*=M^*=1, C^*=0$) identically for both models, which identifies the failure as structural rather than model-specific.

The Contract on Third-Party Agent Stacks

The strongest external-validity test of an evaluation protocol is its application to systems built by others. The contract has two layers: an *invariant core* — four event roles (a memory

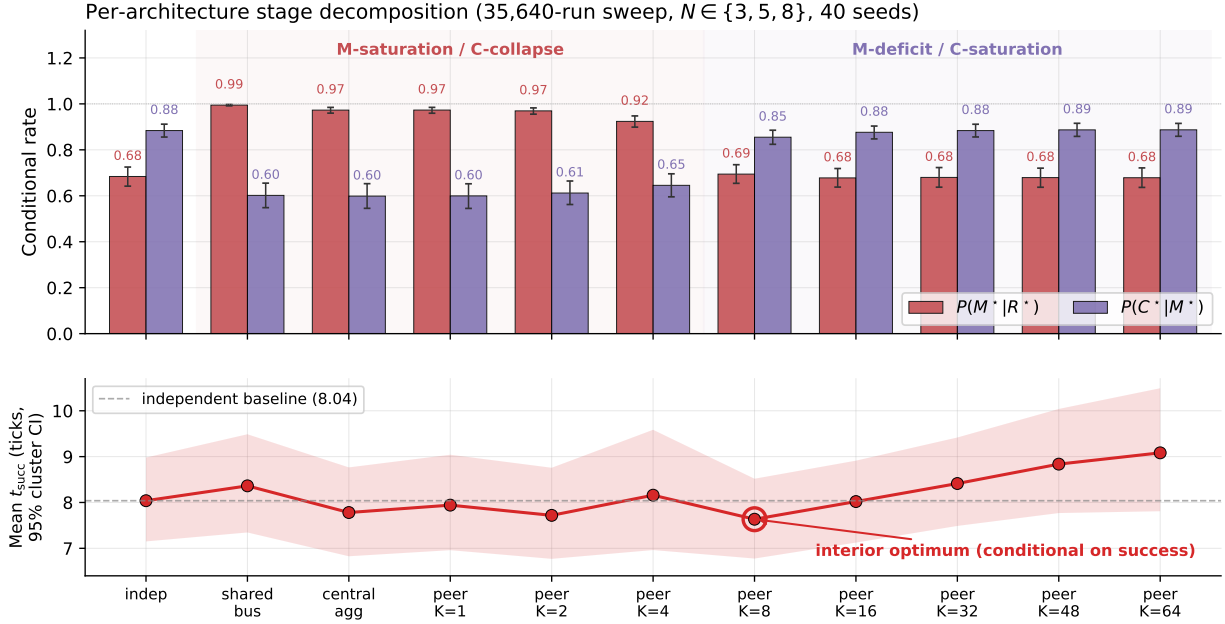


Figure 2: Per-architecture conditional Q/R/M/C rates (top) and mean time-to-success (bottom) on the 35,640-run sweep. Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) sit in the M-saturation/C-collapse regime; independent and slow peer ($K \geq 16$) in the reverse. Error bars and band: 95% cluster-bootstrap CIs over the 81 design cells. Time-to-success among successful runs has an interior minimum at $K=8$ (7.63 vs. 8.04 independent); the sharp statement is the cell-paired contrasts in the text, and the censoring-aware counterpart is also given there.

query is issued; returned evidence satisfies the task; the agent commits to a target; the task completes) — and a *system-specific adapter* that maps logged actions to those roles. This section instruments three third-party stacks — an OpenAI-SDK function-calling loop, LangGraph’s ReAct agent, and AutoGen — with one tool-trace adapter shared by all three: an episodic household memory behind `recall/goto/take` tools; Q^* on querying, R^* on attribution (a returned record names the item at its true place), M^* on a commitment (`goto`) to the true place, C^* on completion within a tool budget. A tool trace exposes no lock chain, so these are the *marginal event frequencies* of §: an agent can commit to the true place without a supporting retrieval, the events are not nested, and no factorisation is claimed — the four rates serve as separate diagnostics. What varies across system classes is the adapter, not the event roles; two revisions of this adapter are recorded in the pre-run registration. Four variants engineer one failure each; all predictions were registered before the runs.

Three findings emerge (Table 2, Fig. 4). **(1) Structural localizations are exact and framework-independent.** With no consolidated fact, $R^*=0.00$; with the budget below requirement, $C^*=0.00$ at $Q^*=R^*=1$. Both hold in 20/20 episodes on every stack with no per-framework adaptation. **(2) The clean control separates the stacks, and the stages localize the difference.** Success on identical clean episodes spreads 0.50–0.75. A leaderboard would report a quality difference and stop; the stages show a pathology shared by all three stacks (first-commitment discipline is weak every-

Table 2: Marginal Q/R/M/C event frequencies on third-party stacks (20 episodes/cell; the events are not nested here and no product is claimed). M_1 : first commitment targets the true place.

Stack	Variant	Q^*	R^*	M^*	C^*	M_1
OpenAI SDK	control	1.00	1.00	.85	.75	.25
	consol. gap	1.00	.00	.50	.40	.25
	stale ambig.	1.00	1.00	.90	.30	.25
	tight budget	1.00	1.00	.25	.00	.25
Lang-Graph	control	1.00	1.00	.70	.50	.50
	consol. gap	1.00	.00	.20	.15	.10
	stale ambig.	1.00	1.00	.65	.60	.50
	tight budget	1.00	1.00	.10	.00	.10
Auto-Gen	control	1.00	1.00	.85	.55	.45
	consol. gap	1.00	.00	.30	.20	.15
	stale ambig.	1.00	1.00	.50	.45	.35
	tight budget	1.00	1.00	.15	.00	.15

where, $M_1=0.25$ –0.50: the first `goto` frequently ignores the agent’s own just-retrieved evidence) together with different loop economics: episode-level M recovers to 0.70–0.85 by re-query, and the stacks differ in how much of that survives to completion (C/M from 0.88 down to 0.65). **(3) The stale distractor has a framework-specific cost.** An outdated record captures the first commitment in 0.20–0.40 of ambiguous episodes, and the completion cost varies (OpenAI SDK: 0.75 $\rightarrow$ 0.30; the others absorb the detour). Some

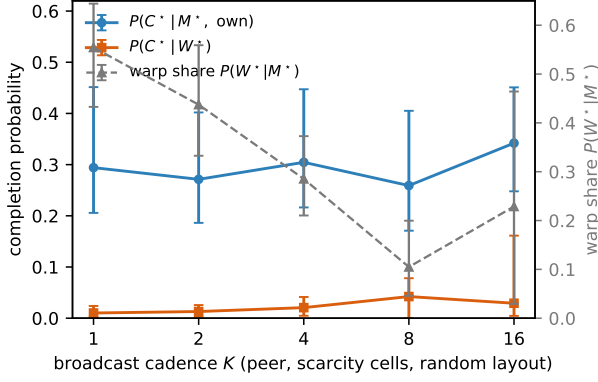


Figure 3: Provenance stratification of $P(C^*|M^*)$ vs. cadence K (peer, scarcity cells, episode-cluster bootstrap CIs). The share of locks backed by foreign evidence (warp share, grey) grows as sharing accelerates, while that stratum’s completion (orange) stays below 0.05 and the own-evidence stratum (blue) stays flat: the C-collapse of fast sharing is concentrated in the foreign-evidence stratum.

registered behavioural predictions assumed a clean control and failed for exactly that reason: the stacks themselves fail a clean memory task 25–50% of the time (finding 2). These are reported as registered failures together with the mechanism.

Model robustness (registered before the runs). Repeating all 240 episodes with a second model (qwen3:4b) confirms the registered prediction exactly: both structural localizations transfer unchanged ($R^*=0.00$ under the consolidation gap and $C^*=0.00$ under the tight budget, 20/20 on every stack). The behavioural profile, by contrast, is a model property that the stages make explicit. qwen3:4b completes every clean episode ($C^*=1.00$ on all three stacks) with first-commitment discipline $M_1=0.90\text{--}0.95$ where llama3.1 managed 0.25–0.50, and is barely distracted by the stale record (0.05–0.30). The exception is AutoGen, which drops qwen’s M_1 to 0.40; the decomposition thus separates the model’s contribution from the effect of the framework’s prompting loop. The full second-model table is in the supplement.

Related Work

Cognitive-map and semantic-map lines motivate explicit place abstraction (Tolman 1948; Gupta et al. 2017; Savinov, Dosovitskiy, and Koltun 2018); goal-conditioned RL conditions on goals or embeddings (Andrychowicz et al. 2017); multi-agent RL studies learned channels (Sukhbaatar, Szlam, and Fergus 2016; Foerster et al. 2018) and gossip protocols (Boyd et al. 2006); LLM-agent systems add explicit memory layers evaluated end-to-end (Wang et al. 2024; Shinn et al. 2023; Packer et al. 2023; Park et al. 2023; Wu et al. 2023). The protocol is orthogonal to these lines: it proposes no architecture, makes any of them stage-diagnosable from logs, and yields a slope-level cadence prediction testable on those logs.

Stage-wise evaluation. The closest line decomposes retrieval-augmented generation: RAGAS scores retrieval and

generation quality separately (Es et al. 2024), RAGChecker refines this to claim-level diagnostics meta-evaluated against human judgments (Ru et al. 2024), and AgentBoard replaces final success with analytic progress rates for multi-turn LLM agents (Ma et al. 2024). Q/R/M/C differs on three axes: its events are logged actions of a task-executing agent, not LLM-judged text quality; it models commitment (M) and the multi-agent coupling channels (shared memory, occupancy) that RAG metrics do not represent; and its diagnoses are validated interventionally (oracle stages; engineered structural faults) rather than by correlation with human ratings. Process mining and conformance checking compare event logs against stage models (van der Aalst 2016; Carmona et al. 2018); the protocol shares that log-level discipline and adds agent-specific event semantics with intervention-based validation. Process reward models score intermediate reasoning steps to supervise training (Lightman et al. 2024); Q/R/M/C is an evaluation-time instrument for memory pipelines.

Limitations

Primary substrates are grid worlds; VMAS is a directional continuous check; the LLM studies use small local models at modest n ; place identity across agents assumes a shared coordinate frame (removed constructively in a companion manuscript); Empirical Claim 1 remains slope-level; a derivation under a generative model would make it a theorem. The four stages are one resolution level: sub-stage causes (accumulation within R; ranking, grounding, and staleness handling within M) require within-link forensics, which the protocol scopes but does not automate. Full tables, proofs of the estimability arguments, assumption stress tests, effect-size methodology, scale-up, and all registered-prediction files are in the supplement.

Conclusion

Q/R/M/C replaces the scalar verdict “the agent’s memory failed” with a diagnosis that names a stage. In our studies this located a consolidation limit in a single agent, a sharing-rate trade-off priced by occupancy in a collective, and a commitment-discipline gap in off-the-shelf agent stacks; in several cases the diagnosis was specific enough to act on. The protocol requires a logging contract rather than architectural modification, and the resulting rates are directly comparable across systems that expose a query/lock interface; for opaque learned policies the protocol measures stage observability itself.

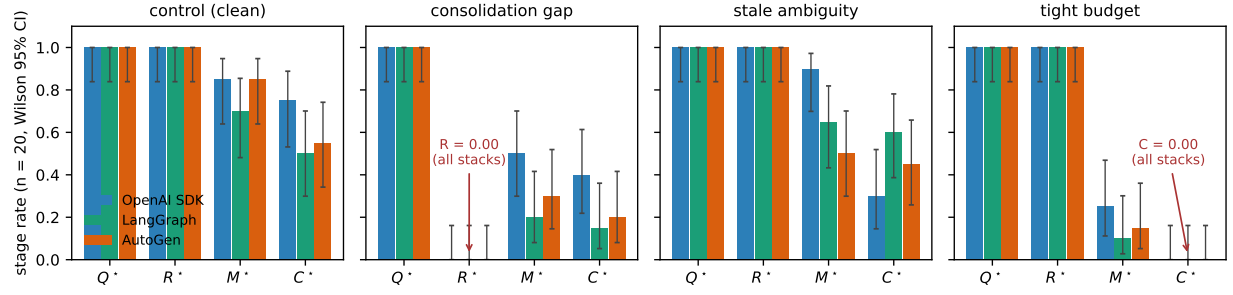


Figure 4: Stage profiles of three third-party stacks on four engineered variants (llama3.1, $n=20$ /cell). Q and R saturate identically; the structural failures give exact zero columns on every stack; the inter-framework spread is confined to M and C.

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- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) **yes**
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) **yes**
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) **yes**
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) **yes**

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) **yes**

If yes, please address the following points:

- 3.2. A motivation is given for why the experi-

ments are conducted on the selected datasets (yes/partial/no/NA) **yes**

- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) **NA**
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **NA**
- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) **yes**
- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) **yes**
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) **NA**

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) **yes**

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) **partial**
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) **yes**
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) **yes**
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) **yes**
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) **partial**
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) **yes**
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of mem-

ory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) [yes](#)

- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) [yes](#)
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) [yes](#)
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) [yes](#)
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) [yes](#)
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) [yes](#)